

## 4.2 Biodiversity

### 4.2.1 Biodiversity

Biodiversity refers to the variety and complexity of life. It is an important indicator in the study of habitats.

Maintaining biodiversity is important for many reasons. Actions to maintain biodiversity must be taken at local, national and global levels.

| Learning outcomes                                                                                                                                                                                                  | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| <i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>(a)</b> how biodiversity may be considered at different levels                                                                                                                                                  | To include: <ul style="list-style-type: none"><li>• habitat biodiversity (sand dunes, woodland, meadows, streams)</li><li>• species biodiversity (species richness and species evenness)</li><li>• genetic biodiversity (different breeds within a species).</li></ul>                                                                                                                                                                                                      |
| <b>(b)</b> <b>(i)</b> how sampling is used in measuring the biodiversity of a habitat and the importance of sampling<br><b>(ii)</b> practical investigations collecting random and non-random samples in the field | <p>To include how sampling can be carried out:</p> <ul style="list-style-type: none"><li>• random sampling</li><li>• non-random sampling (opportunistic, stratified and systematic)</li></ul> <p><b>AND</b><br/>the importance of sampling the range of organisms in a habitat.</p> <p>Techniques to include: quadrats, sweep nets, pitfall traps and pooters.</p> <p><i>M0.2, M1.3, M1.5, M1.4, M1.6, M1.7, M1.9, M1.10, M3.2</i><br/><b>PAG3</b><br/>HSW4, HSW5, HSW6</p> |
| <b>(c)</b> how to measure species richness and species evenness in a habitat                                                                                                                                       | <i>M1.1, M1.5, M2.3, M2.4</i>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>(d)</b> the use and interpretation of Simpson's Index of Diversity ( <i>D</i> ) to calculate the biodiversity of a habitat                                                                                      | <p>The formula will be provided where needed in assessments and does not need to be recalled</p> $D = 1 - \left( \sum \left( \frac{n}{N} \right)^2 \right)$ <p><b>AND</b><br/>the interpretation of both high and low values of Simpson's Index of Diversity (<i>D</i>).</p> <p><i>M1.1, M1.5, M2.3, M2.4</i><br/>HSW5</p>                                                                                                                                                  |

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| <b>(e)</b> how genetic biodiversity may be assessed, including calculations                      | <p>To include calculations of genetic diversity within isolated populations, for example the percentage of gene variants (alleles) in a genome.</p> <p>proportion of polymorphic gene loci = <math display="block">\frac{\text{number of polymorphic gene loci}}{\text{total number of loci}}</math></p> <p>Suitable populations include zoos (captive breeding), rare breeds and pedigree animals.</p> <p><i>M1.1, M1.5, M2.3, M2.4</i><br/>HSW5</p> |
| <b>(f)</b> the factors affecting biodiversity                                                    | <p>To include human population growth, agriculture (monoculture) and climate change.</p> <p><i>M1.3, M1.7, M3.1</i><br/>HSW5, HSW10, HSW12</p>                                                                                                                                                                                                                                                                                                        |
| <b>(g)</b> the ecological, economic and aesthetic reasons for maintaining biodiversity           | <ul style="list-style-type: none"> <li>• Ecological: protecting keystone species (interdependence of organisms) and maintaining genetic resource.</li> <li>• Economic: reducing soil depletion (continuous monoculture).</li> <li>• Aesthetic: protecting landscapes.</li> </ul> <p>HSW12</p>                                                                                                                                                         |
| <b>(h)</b> <i>in situ</i> and <i>ex situ</i> methods of maintaining biodiversity                 | <ul style="list-style-type: none"> <li>• <i>In situ</i> conservation: marine conservation zones and wildlife reserves.</li> <li>• <i>Ex situ</i> conservation: seed banks, botanic gardens and zoos.</li> </ul> <p>HSW7, HSW9, HSW10, HSW12</p>                                                                                                                                                                                                       |
| <b>(i)</b> international and local conservation agreements made to protect species and habitats. | <p>Historic and/or current agreements, including the Convention on International Trade in Endangered Species (CITES), the Rio Convention on Biological Diversity (CBD) and the Countryside Stewardship Scheme (CSS).</p> <p>HSW11, HSW12</p>                                                                                                                                                                                                          |